Curriculum Vitae **October 2026**

Ahmed Elsharkawy

Postdoctoral Fellow

Gwangju, Republic of Korea

✉ elsharkawy@gm.gist.ac.kr

Objective

To contribute to impactful teaching, research, and R&D initiatives by applying my expertise in human-centered design, AI-driven interaction systems, and human-robot interaction. I seek a dynamic academic environment where I can educate and mentor students, drive practical research solutions, collaborate with interdisciplinary teams, and scale technologies for global impact.

Experience

InnoCORE Postdoctoral Fellow / GIST & DGIST **Nov 2025 – Present**

Gwangju Institute of Science and Technology (GIST), College of Artificial Intelligence, Department of AI (Co-affiliated with Daegu Gyeongbuk Institute of Science and Technology (DGIST), InnoCORE Research Group for Bio-Embodied Physical AI), Republic of Korea

- Research Projects
  - Lead Researcher / Human-in-the-Loop Physical AI for XR-Based Digital Therapeutics
  - Research Assistant / Physical AI in Action: Tackling HCI Challenges with XR-Enhanced Robotics for Interpretable, Inclusive, and Immersive Interaction "GIST-MIT Physical AI Research Center"
- Funding
  - InnoCORE program of the Ministry of Science and ICT
  - The National Research Foundation of Korea (NRF) grant funded by the Korean government (MSIT)

Research Assistant Professor / GIST **Oct 2024 – Oct 2025**

GIST, College of Information and Computing, Department of AI Convergence, Human-Centered Intelligent Systems (HCIS) Lab, Republic of Korea

- Research Projects
  - Co-PI / Bridging the gap between reality and virtuality: actuated XR systems powered by sensory intelligence and soft robotics
  - Co-PI / Development of dynamic real-virtual haptic feedback systems using personal robots
- Funding
  - The Basic Science Research Program through the National Research Foundation of Korea (NRF), funded by the Ministry of Education
  - The NRF grant funded by the Korea government (MSIT)

Postdoctoral Fellow / GIST **May 2023 – Sep 2024**

GIST, College of Information and Computing, Department of AI Convergence, Human-Centered Intelligent Systems (HCIS) Lab, Republic of Korea

- Research Projects
  - Project leader / AI-powered intelligent surfaces for adaptive XR environments in vehicles
  - Project leader / Human-AI Collaboration for XR Digitization of Physical Experience
  - Research Assistant / HCI + AI for Human-Centered Physical System Design
- Funding
  - AI-based GIST Research Scientist Project, GIST Institute for Artificial Intelligence (GIAI)
  - GIST-MIT Research Collaboration grant funded by the GIST

Additional Responsibilities across GIST and DGIST Appointments **May 2023 – Present**

- Leading and contributing to multiple studies for SCI(E) journals and top-tier conferences
- Mentoring junior researchers and interns, guiding team integration and idea development
- Supporting teaching for graduate-level Human-Computer Interaction courses

Teaching Assistant / Menoufia University **March 2014 – Aug 2015**

Faculty of Electronic Engineering, Department of Industrial Electronics & Control Engineering, Egypt

- Assisted in teaching undergraduate courses and practical laboratory sessions within the Industrial Electronics & Control Engineering curriculum

GST /ADVANTECH | Automation Engineer (internship) **Jul 2012 – Dec 2012**
Grand Solution for Technology (GST) acted as the local representative of ADVANTECH (Taiwan) in Egypt

- I trained in data acquisition systems and contributed to developing engineering solutions for civil and industrial applications
- Gained experience in system integration and automation

Education

Ph.D. in Integrated Technology (Intelligent Robotics Program) / **GIST** **Sep 2017 – Feb 2023**
Gwangju Institute of Science and Technology (GIST), Gwangju, Republic of Korea

- Ph.D. Dissertation - Robots for Constructing Multitype Real-World Datasets and Augmenting Indoor Spaces: Using RGB-D Camera and Ultra-Wideband Communication for Intuitive Human–Robot Interaction
    - Research Projects
        - Research Assistant / Robots for Cognitive Improvement in Dementia
        - Research Assistant / Lighter-than-air Aerial Robot for Indoor Performances
        - Research Assistant / Multi-dimensional Filming for Immersive Performances
        - Research Assistant / AI-based Interactive Media for Cultural Heritage
    - Funding agencies
        - The Ministry of Trade, Industry & Energy (MOTIE) of Korea; Asia Culture Institute, Asia Culture Center (ACC); Ministry of Culture, Sports and Tourism (MCST) and Korea; Korea Creative Content Agency (KOCCA); National Research Foundation (NRF) of Korea funded by the Korea government (MSIT); and Open AI Dataset Project (AI-Hub, Korea)

MSc in Mechatronics Engineering / **GIST** **Sep 2015 – Aug 2017**
GIST, Gwangju, Republic of Korea

- MSc Dissertation - Design, Control and Evaluation of a Torque-Haptic Feedback Device in an Encounter Type Haptic Display System
    - Research Project
        - Research Assistant / Virtual Plant Training System
    - Funding
        - Ministry of Land, Infrastructure and Transport (MOLIT) of Korea government and Korea Agency for Infrastructure Technology Advancement (KAIA)

BSc in Electronic Engineering / **Menoufia University** **Sep 2007 – Jul 2012**
Faculty of Electronic Engineering, Department of Industrial Electronics & Control Engineering, Egypt

Selected Publications

*** Equal contribution**

- [c.14] **Elsharkawy, A.,** Gim, B., Ataya, A., Kim, S. (2026). *SelfBlending: Artificial Intelligence-driven Augmentation with Hand Interactions for Seamless Reality Blending in Virtual Environments*. in IEEE Transactions on Visualization and Computer Graphics (IEEE TVCG). [Oral Presentation @ IEEE ISMAR 2026]
- [c.13] Gim, B., Kim, G., Yeo, D., Kang, Y., **Elsharkawy, A.**, & Kim, S. (2026). *From Disruption to Immersion: Reimagining Vehicle Motion as Environmental Feedback in In-Car VR*. In Proceedings of the Conference on Human Factors in Computing Systems (ACM CHI).
- [c.12] Ataya, A., **Elsharkawy, A.***, Lee, J., Hwang, S., Seong, M., & Kim, S. (2025). *ReD Shoes: Actuated Footwear for Multisensory Redirected Walking in Virtual Reality*. In Virtual Reality Springer.
- [c.11] Yeo, D., Kim, G., Oh, M., Park, J., Gim, B., Kang, S., **Elsharkawy, A,** & Kim, S. (2025). *AttraCar: Multisensory In-Car VR with Thermal, Airflow, and Motion Feedback through Built-In Vehicle Systems*. In Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (ACM UIST). 🎖️👏
- [c.10] **Elsharkawy, A.**, Ataya, A., Yeo, D., An, E., Hwang, S., & Kim, S. (2024, May). *SYNC-VR: Synchronizing Your Senses to Conquer Motion Sickness for Enriching In-Vehicle Virtual Reality*. In Proceedings of the Conference on Human Factors in Computing Systems (ACM CHI). 🏆
- [c.9] Hwang, S., Oh, J., Kang, S., Seong, M., **Elsharkawy, A.**, & Kim, S. (2024, May). *Ergopulse: Electrifying your lower body with biomechanical simulation-based electrical muscle stimulation haptic*

system in virtual reality. In Proceedings of the Conference on Human Factors in Computing Systems (ACM CHI). 🏆

- [c.8] Kang, S., Kim, G., Hwang, S., Park, J., **Elsharkawy, A**., & Kim, S. (2024, October). *Flip-Pelt: Motor-Driven Peltier Elements for Rapid Thermal Stimulation and Congruent Pressure Feedback in Virtual Reality*. In Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (ACM UIST).
- [c.7] **Elsharkawy, A.**, & Kim, M. S. (2022). *Human–Robot Labeling Framework to Construct Multitype Real-World Datasets*. IEEE Access, 10, 131166-131180.
- [c.6] Naheem, K., **Elsharkawy, A.***, Koo, D., Lee, Y., & Kim, M. (2022). *A UWB-based lighter-than-air indoor robot for user-centered interactive applications*. Sensors, 22(6), 2093.
- [c.5] **Elsharkawy, A.**, Naheem, K., Koo, D., & Kim, M. S. (2021). *A UWB-driven self-actuated projector platform for interactive augmented reality applications*. Applied Sciences, 11(6), 2871.
- [c.4] Jeon, S., **Elsharkawy, A.**, & Kim, M. S. (2021). *Lipreading architecture based on multiple convolutional neural networks for sentence-level visual speech recognition*. Sensors, 22(1), 72.
- [c.3] Ataya, A., Kim, W., **Elsharkawy, A.**, & Kim, S. (2021). *How to interact with a fully autonomous vehicle: naturalistic ways for drivers to intervene in the vehicle system while performing non-driving related tasks*. Sensors, 21(6), 2206.
- [c.2] Ataya, A., Kim, W., **Elsharkawy, A.**, & Kim, S. (2020). *Gaze-head input: Examining potential interaction with immediate experience sampling in an autonomous vehicle*. Applied Sciences, 10(24), 9011.
- [c.1] **Elsharkawy, A.**, Naheem, K., Lee, Y., Koo, D., & Kim, M. S. (2020, September). *LAIDR: A robotics research platform for entertainment applications*. In 2020 IEEE International Conference on Unmanned Aircraft Systems (ICUAS) (pp. 534-539).

Selected Manuscripts in Progress "Anonymized"

(Titles are anonymized and full author lists are withheld for peer-review; I am the first author on the following paper)

- [c.16] (2027). AI-Mediated Context-Aware Interfaces for Room-Scale XR Environments. In Proceedings of the Conference on Human Factors in Computing Systems (ACM CHI).
- [c.15] (2026). User-configurable, LLM-mediated feedback for reflective posture improvement. In the International Journal of Human-Computer Studies.

Patents

- *Method and System for Providing Virtual Reality-Based Multi-Sensory Content*. Republic of Korea patent, Application No. 10-2026-0092893, filed on May 22, 2026; assignee: Gwangju Institute of Science and Technology (GIST); inventors: SeungJun Kim, Dohyeon Yeo, Seongjun Kang, Gwangbin Kim, Bocheon Gim, Jeongju Park, **Ahmed Elsharkawy**, and Minwoo Oh.
- *Haptic Wearable Device for Tactile Experiences*, Republic of Korea registered patent, Registration No. 10-2946187, Application No. 10-2025-0041073, registered March 26, 2026, published March 31, 2026; assignee: Gwangju Institute of Science and Technology (GIST); inventors: SeungJun Kim, Seongjun Kang, Gwangbin Kim, Seokhyun Hwang, Jeongju Park, **and Ahmed Elsharkawy**.
- *Multisensory In-Car VR with Thermal, Airflow, and Motion Feedback through Built-In Vehicle Systems*. Republic of Korea patent, Application No. 10-2025-0157134, filed on October 27, 2025 (Patent Application Filed); applicant: Gwangju Institute of Science and Technology (GIST); inventors: SeungJun Kim, Dohyeon Yeo, Seongjun Kang, Gwangbin Kim, Bocheon Gim, Jeongju Park, **Ahmed Elsharkawy**, and Minwoo Oh.

Teaching courses and labs

- Human-Computer Interaction course [RT5301-01, AI5604-01] / GIST (Co-lecturer / Spring 2025, and Fall 2026)
- Research Methods in Human-Computer Interaction course [AI6317-01, RT5302-01] / GIST (Co-lecturer / Spring 2026 and Fall 2024; Teaching Assistant / Fall 2018)
- HCI+AI research trends and technical analysis course [RT6303-01] / GIST (Teaching Assistant (Course Assessment) / Fall 2025)
- Labs for Power Electronics and Supervisory Control and Data Acquisition (SCADA) / Faculty of Electronic Engineering (Instructor / Spring 2015)
- Labs for Electrical Machines and Control Systems / Faculty of Electronic Engineering (Instructor / Fall 2014)
- Lab for Microcontrollers and Embedded Systems / Faculty of Electronic Engineering (Instructor / Spring 2014)

Programming Languages and Engineering Software

- Python
- C++, embedded systems
- C#, Unity 3D
- Robot Operating System (ROS)
- MATLAB & Simulink
- LabVIEW
- Ladder programming language, PLC (programmable logic controller)
- Electrical circuit design software (Eagle, Proteus, and Altium)

Honors and Awards

InnoCORE Postdoctoral Fellowship **Nov 2025**
Selected for the prestigious Global InnoCORE program (Supported by the Ministry of Science and ICT, Republic of Korea)

Demo Honorable Mention Award & People's Choice Best Demo Award **Sep 2025**
ACM Symposium on User Interface Software and Technology (ACM UIST) [c.11]

Best Paper Honorable Mention **April 2024**
ACM conference on human factors in computing systems (CHI) 2024 [c.10]

Best Paper Honorable Mention **April 2024**
ACM conference on human factors in computing systems (CHI) 2024 [c.9]

Excellence award for the outstanding performance at the AI-based GIST Research Scientist Project **Dec 2023**
Artificial Intelligent school at Gwangju Institute of Science and Technology

Fully funded scholarship for doctoral studies **Aug 2017**
Republic of Korea (government/GIST) scholarship

Fully funded scholarship for master's studies **Aug 2015**
Republic of Korea (government/GIST) scholarship

Academic Service

- Organizer and Session Chair, XRWay'26: The 1st International Workshop on XR on the Way (XRWay): XR for Walking, Driving, and Everyday Mobility @ IEEE ISMAR 2026
- Reviewer, ACM Conference on Human Factors in Computing Systems (CHI)
- Reviewer, International Journal of Human-Computer Interaction
- Reviewer, IEEE International Symposium on Mixed and Augmented Reality (ISMAR)
- Reviewer, Sensors journal, MDPI

Professional Networking and Academic Profiles

- **Personal Website**: Link
- **LinkedIn**: Link
- **Google Scholar**: Link

Media and Press Interview

- SelfBlending VR Tech Integrates Real Objects, Preserves Immersion. Press link: Press Release
- Full immersion in VR inside a moving car… GIST unveils new technology using vehicle motion. Press link: Press Release
- Walking Without Real-World Spatial Constraint, GIST Develops 'Red Shoes' to Overcome VR Walking Limitations Press link: Press Release
- Enjoying 4D VR in a Moving Car, GIST Develops Platform Enabling Immersive Experiences Using Only Built-in Vehicle Systems Press link: Press Release
- dmc channel, 'Logarithm' TV show: An overview of my research journey and discussion about robotics development Episode link: YouTube

- Two GIST doctoral researchers selected for the Ministry of Education's Academic Next Generation Support Project Press link: Press Release
- Human-Centered XR Experiences Break the Boundary between Reality and Virtuality Press link: Press Release
- GIST-MIT continues research results on 'human-centered' AI system created through close collaboration Press link: Press Release